Classical Field Theory — Exercise Sheet 11

Constraints and boundaries

Solutions

Use Lecture 11's canonical bracket, with $\mathbf{\Pi} = -\mathbf{E}$. A primary constraint is imposed only after forming the Hamilton equations. All field data below are real and smooth. The second problem uses Schwartz data on $\mathbb{R}^3$; the third explicitly changes the spatial region and its allowed boundary variations.

Exercise 11.1 (An entire constraint chain). Consider two mechanical coordinates $q(t)$ and $u(t)$ with Lagrangian

$$L(q, u, \dot{q}, \dot{u}) = \frac{1}{2}(\dot{q} - u)^2.$$

Use the following consistency procedure: find the image of the Legendre map, form the canonical Hamiltonian on that image, add an arbitrary multiplier for each primary constraint, and preserve the constraints until no new condition appears. The bracket on $(q, u; p_q, p_u)$ is the ordinary canonical bracket of mechanics.

(a) Carry out every step. State which conditions hold strongly and which hold only on the constraint surface. Classify the constraints.

(b) Find a time-dependent generator producing $\delta q = \epsilon(t)$ and $\delta u = \dot{\epsilon}(t)$ for an arbitrary smooth real function ϵ. Determine the transformation of the total-Hamiltonian multiplier needed to map solutions to solutions.

(c) At a fixed time, show that every two points on the constraint surface are gauge equivalent. How many canonical pairs remain on this quotient?

Solution. For the first part, the Legendre map gives

$$p_q = \dot{q} - u, \qquad p_u = 0.$$

Its velocity Hessian is $\mathrm{diag}(1, 0)$. The primary surface is $p_u = 0$, while $\dot{q} = p_q + u$ is determined. The subtraction gives

$$H_c = p_q(p_q + u) - \frac{1}{2}p_q^2 = \frac{1}{2}p_q^2 + up_q, \qquad H_T = H_c + \lambda p_u.$$

The Hamilton equations on the extended space are

$$\dot{q} = p_q + u, \qquad \dot{u} = \lambda, \qquad \dot{p}_q = 0, \qquad \dot{p}_u = -p_q.$$

Thus preserving $p_u = 0$ requires the secondary constraint $p_q = 0$. Its preservation gives no further condition and leaves λ arbitrary. The mutual constraint brackets vanish strongly. In contrast, $\{p_u, H_T\} = -p_q$ is only weakly zero. Both constraints are first class.

For the second part, define

$$G_\epsilon = \epsilon p_q + \dot{\epsilon} p_u.$$

Taking brackets in the order $\delta F = \{F, G_\epsilon\}$ gives

$$\delta q = \epsilon, \qquad \delta u = \dot{\epsilon}, \qquad \delta p_q = \delta p_u = 0.$$

The combination $\dot q - u$ is unchanged. To check Hamiltonian solutions, put $q' = q + \epsilon$ and $u' = u + \dot\epsilon$. Then

$$\dot q' = p_q + u', \qquad \dot u' = \lambda + \ddot\epsilon.$$

Hence $\lambda' = \lambda + \ddot\epsilon$ preserves every Hamilton equation. A generator containing only the secondary constraint would miss the shift of the auxiliary coordinate.

For the final part, the constraint surface has coordinates (q, u). At any fixed time, choose ϵ and $\dot\epsilon$ equal to the two differences between prescribed points. A smooth function with these two values exists, for example an affine function near that time. Its gauge transformation maps one point to the other. The quotient is a point and has no remaining canonical pair. This conclusion was established directly, without assuming a general constraint-counting theorem.

Exercise 11.2 (A mass term changes the constraint class). On $\mathbb{R}^{1+3}$ take the vacuum density

$$L_M = \frac{1}{2}|\mathbf{A}_t + \nabla\phi_{\mathrm{em}}|^2 - \frac{1}{2}|\operatorname{curl}\mathbf{A}|^2 + \frac{M^2}{2}(\phi_{\mathrm{em}}^2 - |\mathbf{A}|^2), \qquad M > 0.$$

At each time all canonical fields and the multiplier are Schwartz. Use the extended bracket of Lecture 11 and define $P[\eta] = \int \eta\pi_\phi \,\mathrm{d}^3x$ for real Schwartz η.

(a) Derive the momenta, canonical Hamiltonian and consistency chain. Does preserving the secondary constraint leave $\partial_t\phi_{\mathrm{em}}$ arbitrary?

(b) Put $\mathcal{D}_M = \operatorname{div}\mathbf{\Pi} - M^2\phi_{\mathrm{em}}$ and $C_M[\chi] = -\int \chi\mathcal{D}_M \,\mathrm{d}^3x$. Compute all primary/secondary brackets. For smearings with $\int \eta\chi = 1$, write the two-by-two off-diagonal bracket block and its inverse. Why does its inverse have no massless limit?

(c) Impose both constraints and express the scalar variables in terms of $\mathbf{A}$ and $\mathbf{\Pi}$. Pull back the canonical form and determine how many spatial component pairs remain. Contrast this with Maxwell's gauge quotient.

Solution. For the first part, differentiation in the velocities gives

$$\pi_\phi = 0, \qquad \mathbf{\Pi} = \mathbf{A}_t + \nabla\phi_{\mathrm{em}} = -\mathbf{E}.$$

The Legendre subtraction, with $\mathbf{A}_t = \mathbf{\Pi} - \nabla\phi_{\mathrm{em}}$, is

$$H_c = \int \left[\frac{1}{2}(|\mathbf{\Pi}|^2 + |\operatorname{curl}\mathbf{A}|^2) - \mathbf{\Pi}\cdot\nabla\phi_{\mathrm{em}} - \frac{M^2}{2}\phi_{\mathrm{em}}^2 + \frac{M^2}{2}|\mathbf{A}|^2\right]\mathrm{d}^3x.$$

On a cube Q_R, the term containing ϕ_{em} has the boundary contribution

$$-\int_{Q_R} \mathbf{\Pi}\cdot\nabla\phi_{\mathrm{em}} = \int_{Q_R} \phi_{\mathrm{em}} \operatorname{div}\mathbf{\Pi} - \int_{\partial Q_R} \phi_{\mathrm{em}}\mathbf{\Pi}\cdot n \,\mathrm{d}S.$$

Schwartz decay removes the face integral as $R \to \infty$. The magnetic variation has the same vanishing boundary term as in Lecture 11. Adding $\int \lambda\pi_\phi$ gives

$$\partial_t\phi_{\mathrm{em}} = \lambda, \qquad (\pi_\phi)_t = -\operatorname{div}\mathbf{\Pi} + M^2\phi_{\mathrm{em}} = -\mathcal{D}_M,$$
$$\mathbf{A}_t = \mathbf{\Pi} - \nabla\phi_{\mathrm{em}}, \qquad \mathbf{\Pi}_t = -\operatorname{curl}\operatorname{curl}\mathbf{A} - M^2\mathbf{A}.$$

The primary condition therefore requires $\mathcal{D}_M = 0$. Its derivative is

$$(\mathcal{D}_M)_t = -M^2 \operatorname{div}\mathbf{A} - M^2\lambda.$$

Since $M > 0$, preservation fixes $\lambda = -\operatorname{div}\mathbf{A}$. There is no tertiary constraint. The Lorenz condition is now forced by consistency, rather than selectable by residual Maxwell gauge freedom.

For the second part, the nonzero functional derivatives are

$$P_{\pi_\phi} = \eta, \qquad (C_M)_{\phi_{\mathrm{em}}} = M^2\chi, \qquad (C_M)_{\mathbf{\Pi}} = \nabla\chi.$$

They give

$$\{P[\eta], P[\zeta]\} = 0, \quad \{C_M[\chi], C_M[\vartheta]\} = 0, \quad \{P[\eta], C_M[\chi]\} = -M^2 \int \eta\chi \, \mathrm{d}^3x.$$

Thus the constraints do not form Maxwell's first-class algebra. The normalized scalar bracket block and inverse are

$$\begin{pmatrix} 0 & -M^2 \\ M^2 & 0 \end{pmatrix}, \qquad \begin{pmatrix} 0 & M^{-2} \\ -M^{-2} & 0 \end{pmatrix}.$$

This is an invertible second-class block for positive M; it is singular at $M = 0$. These displayed matrices describe the scalar constraint pairing, not an unqualified inverse of a pointwise field-distribution matrix.

For the final part, the constraints solve the scalar variables as

$$\pi_\phi = 0, \qquad \phi_{\mathrm{em}} = M^{-2} \operatorname{div}\mathbf{\Pi}.$$

Every Schwartz spatial pair gives such scalar data. On this constraint space, $\delta\pi_\phi = 0$, so the canonical form restricts to

$$\Omega_M = \int_{\mathbb{R}^3} \sum_{i=1}^{3} \delta A_i \wedge \delta\Pi_i \, \mathrm{d}^3x.$$

It is nondegenerate on the three spatial Schwartz pairs, by testing a vector (a, p) against $(-p, a)$. There is no spatial gradient null direction to quotient. Maxwell instead has a longitudinal constraint on $\mathbf{\Pi}$ and a spatial gauge identification of $\mathbf{A}$. The positive-mass theory therefore has three component pairs rather than two transverse pairs.

Exercise 11.3 (A boundary generator need not vanish). Let $\Omega = B_1(0) \subset \mathbb{R}^3$. For this problem the phase variables $\mathbf{A}, \mathbf{\Pi}$ are smooth on a neighbourhood of $\overline{\Omega}$, with no fixed normal trace for $\mathbf{\Pi}$. Their canonical form is $\int_\Omega \delta A_i \wedge \delta\Pi_i \, \mathrm{d}^3x$. A functional has a *smooth bulk derivative* if its variation is an integral of smooth coefficients against arbitrary smooth field variations, without a boundary evaluation term. For a fixed smooth real χ, set

$$C_\Omega[\chi] = -\int_\Omega \chi \operatorname{div}\mathbf{\Pi} \, \mathrm{d}^3x, \qquad G_\Omega[\chi] = \int_\Omega \mathbf{\Pi} \cdot \nabla\chi \, \mathrm{d}^3x.$$

(a) Derive their relation, including the surface integral, and vary it. Explain why $C_\Omega[\chi]$ generally lacks a smooth bulk derivative when χ is nonzero on the boundary, while $G_\Omega[\chi]$ has one.

(b) Find the Hamiltonian vector of $G_\Omega[\chi]$. On the Gauss surface $\operatorname{div}\mathbf{\Pi} = 0$, must its value and its contraction with the constrained canonical form vanish?

(c) Evaluate both functionals and their boundary difference for $\mathbf{\Pi} = e_1$ and $\chi = x^1$. Identify the boundary assumption that distinguishes this problem from the all-space integrations by parts in Lecture 11.

Solution. For the first part, the product rule and the ball divergence formula give

$$C_\Omega = G_\Omega - \int_{\partial\Omega} \chi\mathbf{\Pi} \cdot n \, \mathrm{d}S.$$

The domain, χ and normal are fixed during phase variation. Hence

$$\delta C_\Omega = \int_\Omega \nabla\chi \cdot \delta\mathbf{\Pi}\, \mathrm{d}^3x - \int_{\partial\Omega} \chi\delta\mathbf{\Pi}\cdot n\, \mathrm{d}S, \qquad \delta G_\Omega = \int_\Omega \nabla\chi\cdot\delta\mathbf{\Pi}\, \mathrm{d}^3x.$$

Interior tests force any proposed smooth momentum derivative of C_Ω to equal $\nabla\chi$. It cannot then account for the remaining boundary term. If χ has nonzero trace, choose a smooth variation with normal trace $\delta\mathbf{\Pi}\cdot n = \chi$ there. Its boundary term is $-\int_{\partial\Omega}\chi^2 \neq 0$. Thus no smooth bulk derivative exists. Adding the displayed boundary functional cancels this obstruction exactly.

For the second part, contraction with the canonical form shows

$$X_{G_\Omega} = (\nabla\chi, 0), \qquad \Omega(X_{G_\Omega}, (a,p)) = \int_\Omega \nabla\chi\cdot p\, \mathrm{d}^3x.$$

For a tangent with $\operatorname{div} p = 0$, this becomes $\int_{\partial\Omega}\chi p\cdot n\, \mathrm{d}S$. It need not be zero. On the Gauss surface $C_\Omega = 0$, but $G_\Omega = \int_{\partial\Omega}\chi\mathbf{\Pi}\cdot n\, \mathrm{d}S$ can remain nonzero.

For the last part, $\operatorname{div} e_1 = 0$ and $\nabla x^1 = e_1$, so

$$C_\Omega = 0, \qquad G_\Omega = \operatorname{vol}(B_1) = \frac{4\pi}{3}.$$

On the unit sphere $x^1 = n_1$. Rotation to its polar axis gives

$$\int_{\partial\Omega} x^1 n_1\, \mathrm{d}S = 2\pi\int_{-1}^{1} s^2 \mathrm{d}s = \frac{4\pi}{3}.$$

The same value is the pairing with the transverse tangent $(0, e_1)$. These boundary transformations are therefore not null directions of this constrained form. The all-space Schwartz integrations by parts in Lecture 11 have vanishing surface contributions. They do not justify dropping a free boundary trace on this ball. A gauge reduction must specify which boundary transformations it identifies.